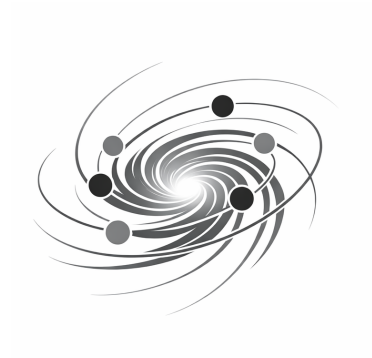


Part of the Monistic Continuum Model (MCM)



Dark Matter (DM)

Emergent Unbound Geometry within the Monistic Continuum

DOI: [10.5281/zenodo.22678381](https://doi.org/10.5281/zenodo.22678381)

Version 02 — 9. November 2026

© 2026 Walter Moosbrugger

Licensed under CC BY 4.0

Abstract

The Monistic Continuum Model (MCM) describes a single continuous medium without physical space, without physical time, without fields, without particles, and without forces. All observable phenomena arise from internal geometric relations of the medium, defined through stress, rotation, curvature, and update dynamics.

In its pre-spatial state, the medium contains only unbound geometry. This geometry is fully described by the spaceless state mapping

$$C(\alpha, \Omega) = (\rho(\alpha, \Omega), v_A(\alpha, \Omega), \sigma_{AB}(\alpha, \Omega), \tau(\alpha, \Omega))$$

Stress generates internal geometry, rotation modulates this geometry, and curvature emerges as pure medium curvature from $\gamma_{AB}(\alpha, \Omega)$. The update rate $\tau(\alpha, \Omega)$ governs the internal reactivity of the medium and replaces any notion of physical time.

Bound geometry arises precisely when the internal update

$$DC(\alpha, \Omega) \approx 0$$

reaches local stability. This stable bound geometry manifests as vortices and forms the foundation of matter and emergent space. Space is not a fundamental entity but the external appearance of bound stress-rotation-curvature structures:

$$g_{ij}(x, \Omega) = \gamma_{AB}(\alpha, \Omega) \frac{\partial X^A}{\partial x^i} \frac{\partial X^B}{\partial x^j}.$$

Matter is bound space. Unbound geometry forms the pre-spatial background from which all bound structures emerge.

This work formulates the complete ontological and axiometric foundation of the MCM and demonstrates how all observable phenomena — including bound structures, rotational-geometric modes, and large-scale curvature domains — arise from a single medium.

Contents

1	Fundamental Axioms of the Medium	9
2	Formal Definitions	12
3	MCM Formal Structure	14
4	Operator Algebra	17
5	Functional Spaces	19
6	Unbound Geometry	22
7	DM Definition	25
8	Residual Stress Density	28
9	Dissolution Background	30
10	Stress–Energy Tensor	32
11	Spectral Operators	34
12	Cosmological Implications	36
13	Gravitational Signatures	38
14	Spectral Signatures	40
15	Predictions	42
16	Experimental Approaches	45
17	Conclusion	47
18	Appendix A: Formal Mathematical Details	49
19	References	52
20	Citation Recommendation	56

Summary of Main Results

This work formulates the complete ontological and axiometric structure of the Monistic Continuum Model (MCM). All results arise solely from the fundamental axioms of the medium and the pre-spatial state mapping $C(\alpha, \Omega)$. The main findings can be summarized in five points.

Statement of the Main Theorem

The central result is:

Bound geometry arises precisely when $DC(\alpha, \Omega) \approx 0$.

From this it follows directly:

Matter is bound geometry.

and

Space is the external appearance of bound geometry.

This theorem links the internal update dynamics of the medium to the emergence of stable vortex structures, which appear as bound stress–rotation–curvature geometry.

Operatorial Innovations

The work introduces three central geometric constructions:

1. ****Pre-spatial geometry****

$$\gamma_{AB}(\alpha, \Omega) = \delta_{AB} + \alpha_1 \sigma_{AB} + \alpha_2 I_1 \delta_{AB},$$

which arises solely from stress.

2. ****Internal curvature****

$$\mathcal{R}_{ABCD} = R(\gamma_{AB}),$$

which is pure medium curvature.

3. ****Rotational geometry****

$$L_{AB} = \frac{1}{2}(\nabla_A v_B - \nabla_B v_A),$$

which generates pre-spatial wave-like patterns.

These three constructions form the foundation of all emergent phenomena.

Mathematical Consequences

From the axioms, the following mathematical consequences arise:

- **Unbound geometry**

$$DC(\alpha, \Omega) \neq 0$$

is the pre-spatial form of the medium.

- **Temporarily bound geometry**

$$DC(\alpha, \Omega) \approx 0 \quad (\text{temporary})$$

forms transitional structures.

- **Stable bound geometry (vortices)**

$$DC(\alpha, \Omega) = 0$$

produces persistent bound stress geometry.

- **Emergence of space**

$$g_{ij}(x, \Omega) = \gamma_{AB} \frac{\partial X^A}{\partial x^i} \frac{\partial X^B}{\partial x^j}.$$

Space is therefore a mathematical projection of bound geometry.

Physical Implications (Emergent Geometry)

Since the MCM contains no physical entities, all “physical” appearances arise solely from bound geometry:

- bound stress geometry appears as matter, - rotational geometric modes appear as wave-like patterns, - curvature domains appear as large-scale structure, - unbound geometry forms the pre-spatial background.

All observable phenomena are projections of the internal medium geometry.

Comparison with Λ CDM and GR

The comparison level is purely geometric:

- In Λ CDM and GR, space is fundamental. In MCM, space is **emergent**.

- In Λ CDM and GR, matter is fundamental. In MCM, matter is **bound geometry**.
- In Λ CDM and GR, fields and forces exist. In MCM, **only stress, rotation, and curvature** exist.
- In Λ CDM and GR, time exists. In MCM, **only the update rate** $\tau(\alpha, \Omega)$ exists.

Thus, the MCM provides a fully geometric alternative to spacetime-based models.

Notation

The notation of the Monistic Continuum Model (MCM) is entirely pre-spatial and ontologically minimal. It describes only internal relations of the medium and avoids any form of physical or spacetime structure. All symbols refer to the medium, its internal geometry, and its update dynamics.

Sets, Spaces, Operators

- M — the medium; the single ontological entity.
- α — medium index; not a spatial coordinate.
- Ω — global update index; not physical time.
- $C(\alpha, \Omega)$ — pre-spatial state mapping of the medium.
- $\mathcal{G}$ — set of all pre-spatial geometries.
- $\mathcal{U}$ — set of unbound geometries.
- $\mathcal{B}$ — set of bound geometries (vortices).
- DC — internal update derivative.
- $R(\alpha, \Omega)$ — residual stress function.

All sets are internal classifications of the medium and possess no topological or metric structure.

Index Conventions

- Latin indices $A, B, C, \dots$ denote internal geometric directions of the medium (not spatial directions).
- Greek indices $\alpha, \beta, \dots$ denote medium indices (not coordinates).
- The update index Ω is a global reactivity parameter (not time).

No coordinate basis and no metric exist before bound geometry emerges.

Functional Notation

- $\rho(\alpha, \Omega)$ — geometric density (not mass).
- $v_A(\alpha, \Omega)$ — geometric motion (not velocity).
- $\sigma_{AB}(\alpha, \Omega)$ — stress geometry (central ontological quantity).
- $\tau(\alpha, \Omega)$ — update rate (not time).
- $I_1(\alpha, \Omega) = \text{tr}(\sigma)$ — stress invariant.
- $\gamma_{AB}(\alpha, \Omega)$ — internal pre-spatial geometry.
- $\mathcal{R}_{ABCD}(\alpha, \Omega)$ — internal curvature of the medium.
- $L_{AB}(\alpha, \Omega)$ — rotational geometric quantity.

All functions are internal geometric mappings of the medium.

Tensor Notation

The tensors of the MCM are not physical tensors but purely geometric relations of the medium:

- σ_{AB} — stress tensor of the medium.
- γ_{AB} — pre-spatial geometry.
- $\mathcal{R}_{ABCD}$ — internal curvature.
- L_{AB} — rotational geometric tensor.

These tensors exist solely within the medium and have no interpretation as spacetime tensors. Only through bound geometry does a projection into an emergent metric arise:

$$g_{ij}(x, \Omega) = \gamma_{AB}(\alpha, \Omega) \frac{\partial X^A}{\partial x^i} \frac{\partial X^B}{\partial x^j}.$$

This is the only point at which tensors acquire an external appearance.

Preliminaries

The following preliminary remarks define the minimal mathematical and geometric foundations of the Monistic Continuum Model (MCM). They are strictly pre-spatial and ontologically closed. No physical spaces, times, fields, or forces exist. All structures arise solely from internal relations of the medium.

Topological Preliminaries

The medium M possesses no topological structure in the physical sense. It is merely a connected ontological entity.

The only admissible “topology” is the internal neighborhood relation:

$$N : M \rightarrow \mathcal{P}(M),$$

which assigns to each medium index α a finite set of indices that interact internally.

This relation is:

- non-spatial,
- non-metric,
- non-coordinate-based,
- purely ontological.

There exists no structure of open sets, no metric, no manifold, and no physical topology.

Functional Analysis Preliminaries

All functions of the MCM are internal geometric mappings of the medium:

$$C(\alpha, \Omega) = \left(\rho(\alpha, \Omega), v_A(\alpha, \Omega), \sigma_{AB}(\alpha, \Omega), \tau(\alpha, \Omega) \right).$$

The function spaces are:

- $\mathcal{F}_\rho$ — space of density mappings,
- $\mathcal{F}_v$ — space of geometric motions,
- $\mathcal{F}_\sigma$ — space of stress geometries,
- $\mathcal{F}_\tau$ — space of update rates.

These spaces possess:

- no norm,
- no topology,
- no metric,
- no physical interpretation.

They are purely classifications of internal medium geometry.

Operator Algebra Preliminaries

The operators of the MCM are not physical operators but internal geometric transformation rules of the medium.

The fundamental operators are:

$$\gamma_{AB} = \delta_{AB} + \alpha_1 \sigma_{AB} + \alpha_2 I_1 \delta_{AB},$$

$$\mathcal{R}_{ABCD} = R(\gamma_{AB}),$$

$$L_{AB} = \frac{1}{2}(\nabla_A v_B - \nabla_B v_A),$$

$$DC = \text{internal update derivative},$$

$$R(\alpha, \Omega) = \mathcal{R}[C, W](\alpha, \Omega).$$

These operators do not form an algebraic structure in the physical sense. They are purely ontological transformation rules of the medium.

There exist:

- no commutators,
- no Lie algebras,
- no groups,
- no physical operator spaces.

All operators are internal geometric relations.

Measure-Theoretic Preliminaries

The medium possesses no measure structure. There is no volume, no length, no area, no integration. The only admissible “measurement” is internal averaging over medium indices:

$$\langle f \rangle_\alpha = \frac{1}{|N(\alpha)|} \sum_{\beta \in N(\alpha)} f(\beta, \Omega),$$

where $N(\alpha)$ is the internal neighborhood relation.

This is:

- not integration,
- not a measure,
- not physical statistics,
- not geometric integration.

It is merely an internal aggregation of pre-spatial geometry.

1 Fundamental Axioms of the Medium

The Monistic Continuum Model (MCM) is based on a minimal set of ontological axioms. These axioms define a single continuous medium without physical space, without physical time, without fields, without particles, and without forces. All observable phenomena arise from internal geometric relations of the medium.

In its pre-spatial state, the medium possesses only *unbound geometry*. Only through stable binding of this geometry does *bound geometry* emerge, which in the MCM is identical to *matter*.

Axiom 1: The Medium

The medium is the single ontological entity. It possesses no spatial or temporal structure. All structures arise from internal stress, rotation, and curvature.

Ontology: Only the medium.

Axiom 2: Pre-Spatial State Mapping

The state of the medium is described by the spaceless mapping

$$C(\alpha, \Omega) = \left(\rho(\alpha, \Omega), v_A(\alpha, \Omega), \sigma_{AB}(\alpha, \Omega), \tau(\alpha, \Omega) \right)$$

where:

- α is a medium index (not a spatial coordinate),
- Ω is a global update index,
- all components are purely geometric quantities.

This mapping fully describes the medium prior to the emergence of space.

Axiom 3: Stress-Induced Internal Geometry

Stress generates internal geometric relations:

$$\gamma_{AB}(\alpha, \Omega) = \delta_{AB} + \alpha_1 \sigma_{AB}(\alpha, \Omega) + \alpha_2 I_1(\alpha, \Omega) \delta_{AB},$$

with

$$I_1(\alpha, \Omega) = \text{tr}(\sigma(\alpha, \Omega)).$$

This geometry is pre-spatial and unbound.

Axiom 4: Internal Curvature

Curvature arises from the internal geometry:

$$\mathcal{R}_{ABCD}(\alpha, \Omega) = R(\gamma_{AB}(\alpha, \Omega)).$$

This is pure medium curvature, not spatial curvature.

Axiom 5: Rotational Behavior

Rotation is an internal geometric quantity:

$$L_{AB}(\alpha, \Omega) = \frac{1}{2}(\nabla_A v_B - \nabla_B v_A).$$

Rotation interacts with stress and curvature and generates pre-spatial wave-like patterns.

Axiom 6: Update Rate

The update rate is an ontological reactivity quantity:

$$\tau(\alpha, \Omega) \in \mathbb{R}_{\geq 0}.$$

It is not time; it modulates the strength of internal updating.

Axiom 7: Internal Updating

The internal evolution of the medium is described by

$$DC(\alpha, \Omega)$$

Bound geometry arises precisely when

$$DC(\alpha, \Omega) \approx 0.$$

Stable bound geometry manifests as vortices.

Axiom 8: Constitutive Dynamics

Stress reorganizes according to

$$D\sigma_{AB}(\alpha, \Omega) = F(Vv, \sigma)_{AB},$$

with internal transport

$$Vv(\sigma_{AB}) = v_C \nabla_C \sigma_{AB}.$$

All dynamics arise from internal geometric relations.

Axiom 9: Emergence of Space from Bound Geometry

Space emerges solely from *bound geometry*. When a vortex becomes stable, a local mapping arises:

$$x = X(\alpha, \Omega),$$

and the emergent metric is

$$g_{ij}(x, \Omega) = \gamma_{AB}(\alpha, \Omega) \frac{\partial X^A}{\partial x^i} \frac{\partial X^B}{\partial x^j}.$$

Space is the external appearance of bound geometry.

Axiom 10: Matter as Bound Space

Matter is not fundamental. Matter is *bound space*. It arises as stable binding of stress, rotation, and curvature:

$$M(x, \Omega) = \text{bound stress--rotation--curvature structures.}$$

The vortex is the only structure that fully satisfies this binding.

Summary

The MCM possesses two levels:

Medium: $C(\alpha, \Omega)$ (unbound geometry),

Matter: $M(x, \Omega)$ (bound geometry = space).

Space emerges from bound geometry. Matter is bound space. All phenomena are emergent geometric effects of a single medium.

2 Formal Definitions

The following definitions specify the fundamental geometric units of the Monistic Continuum Model (MCM). They are entirely pre-spatial and ontologically closed. All definitions refer exclusively to the medium and its internal stress, rotation, and curvature geometry.

Definition: Vortex Cell

A vortex cell is a locally bound configuration of the medium characterized by stable internal geometry. It arises precisely when

$$DC(\alpha, \Omega) = 0.$$

A vortex cell is not a physical structure but a stable bound stress–rotation–curvature geometry of the medium.

The set of all vortex cells is

$$\mathcal{B} = \{\alpha \in M \mid DC(\alpha, \Omega) = 0\}.$$

Definition: Local State

The local state of the medium at a medium index α is the pre-spatial state mapping

$$C(\alpha, \Omega) = (\rho(\alpha, \Omega), v_A(\alpha, \Omega), \sigma_{AB}(\alpha, \Omega), \tau(\alpha, \Omega)).$$

This state is entirely geometric and has no physical interpretation. It describes the internal configuration of the medium prior to the emergence of space.

Definition: Neighborhood Operator N

The neighborhood operator

$$N : M \rightarrow \mathcal{P}(M)$$

assigns to each medium index α a finite set of indices that interact internally.

$$N(\alpha) = \{\beta \in M \mid \beta \text{ influences } \alpha\}.$$

This relation is:

- non-spatial,
- non-metric,
- non-coordinate-based,
- purely ontological.

Definition: Global Configuration Space V

The global configuration space of the medium is the set of all local states:

$$V = \{C(\alpha, \Omega) \mid \alpha \in M\}.$$

This space has:

- no topology,
- no norm,
- no metric,
- no physical structure.

It is a purely geometric classification of internal medium states.

Definition: Stress Field $\sigma : M \rightarrow \mathcal{G}$

The stress field of the medium is the mapping

$$\sigma_{AB} : M \rightarrow \mathcal{G},$$

which assigns to each medium index an internal stress geometry.

Stress is the central ontological quantity of the MCM. All geometry arises from stress.

Definition: Rotation Field $\omega : M \rightarrow \mathcal{G}$

The rotational geometric field is defined by

$$L_{AB}(\alpha, \Omega) = \frac{1}{2}(\nabla_A v_B - \nabla_B v_A).$$

Rotation is not physical rotation but an internal geometric reorganization of the medium. It generates pre-spatial wave-like patterns.

Definition: Curvature Functional $K : M \rightarrow \mathcal{G}$

The internal curvature of the medium arises from the pre-spatial geometry:

$$\mathcal{R}_{ABCD}(\alpha, \Omega) = R(\gamma_{AB}(\alpha, \Omega)),$$

where

$$\gamma_{AB} = \delta_{AB} + \alpha_1 \sigma_{AB} + \alpha_2 I_1 \delta_{AB}.$$

This curvature is pure medium curvature and has no spatial interpretation.

3 MCM Formal Structure

The formal structure of the Monistic Continuum Model (MCM) is based on a single continuous medium whose internal geometry is defined by stress, rotation, curvature, and update dynamics. This structure is entirely pre-spatial and has no physical interpretation. All bound and unbound forms of geometry arise from the following ontological constructions.

Continuum Medium M

The medium M is the single fundamental entity of the MCM. It possesses:

- no spatial structure,
- no temporal structure,
- no physical fields,

- no particles,
- no forces.

All structures arise from the pre-spatial state mapping

$$C(\alpha, \Omega) = \left(\rho(\alpha, \Omega), v_A(\alpha, \Omega), \sigma_{AB}(\alpha, \Omega), \tau(\alpha, \Omega) \right).$$

Stress Field σ

Stress is the central ontological quantity of the medium. The stress field is defined by

$$\sigma_{AB} : M \rightarrow \mathcal{G},$$

and generates internal geometric relations. All geometry in the MCM is stress geometry.

The stress invariant is

$$I_1(\alpha, \Omega) = \text{tr}(\sigma(\alpha, \Omega)).$$

Rotation Field ω

Rotation is an internal geometric reorganization quantity of the medium. It is defined by the rotational geometric tensor

$$L_{AB}(\alpha, \Omega) = \frac{1}{2} \left(\nabla_A v_B - \nabla_B v_A \right).$$

Rotation is not physical rotation but a pre-spatial reorganization pattern that generates wave-like structures.

Curvature Functional K

Curvature arises from the stress-induced internal geometry

$$\gamma_{AB}(\alpha, \Omega) = \delta_{AB} + \alpha_1 \sigma_{AB}(\alpha, \Omega) + \alpha_2 I_1(\alpha, \Omega) \delta_{AB}.$$

The internal curvature is

$$\mathcal{R}_{ABCD}(\alpha, \Omega) = R(\gamma_{AB}(\alpha, \Omega)).$$

This curvature is pure medium curvature and has no spatial interpretation.

Update-Rate Functional T

The update rate is an ontological reactivity quantity of the medium:

$$\tau(\alpha, \Omega) \in \mathbb{R}_{\geq 0}.$$

It governs the internal evolution of the medium and replaces any form of physical time.

The internal update derivative is

$$DC(\alpha, \Omega),$$

and describes the change of the state mapping under updating.

Bound/Unbound Geometry Classification

The geometry of the medium appears in two fundamental forms:

Unbound Geometry

$$DC(\alpha, \Omega) \neq 0.$$

This is the pre-spatial form of the medium. It possesses:

- no metric,
- no orientation,
- no stability.

Unbound geometry forms the background from which bound structures emerge.

Bound Geometry

$$DC(\alpha, \Omega) = 0.$$

Bound geometry is stable stress–rotation–curvature geometry. It manifests as vortices and forms the foundation of matter and emergent space.

The set of bound geometry is

$$\mathcal{B} = \{\alpha \in M \mid DC(\alpha, \Omega) = 0\}.$$

4 Operator Algebra

The operators of the Monistic Continuum Model (MCM) are not physical operators but purely geometric transformation rules of the medium. They act exclusively on the pre-spatial state mapping $C(\alpha, \Omega)$ and generate internal relations between stress, rotation, curvature, and update dynamics. All operators are ontological and pre-spatial; they possess no interpretation in space or time.

Geometric Operator $G : V \rightarrow \Sigma$

The geometric operator G produces the internal pre-spatial geometry from the global configuration space $V = \{C(\alpha, \Omega)\}$:

$$\gamma_{AB}(\alpha, \Omega) = G(C(\alpha, \Omega)) = \delta_{AB} + \alpha_1 \sigma_{AB}(\alpha, \Omega) + \alpha_2 I_1(\alpha, \Omega) \delta_{AB}.$$

G is not a metric construction but a stress-induced geometric mapping of the medium.

Topological Operator $T : V \rightarrow V$

The topological operator T reorganizes the internal neighborhood relation of the medium:

$$T(C(\alpha, \Omega)) = C(\alpha, \Omega) \circ N(\alpha),$$

where $N(\alpha)$ is the internal interaction set.

T does not generate physical topology but a purely ontological interaction structure.

Dynamical Operator $D : V \rightarrow V$

The dynamical operator D describes the internal reorganization of the medium under updating:

$$D(C(\alpha, \Omega)) = DC(\alpha, \Omega).$$

It is not a temporal derivative but an internal geometric update rule.

Stability Operator $S : V \rightarrow \mathbb{R}_{\geq 0}$

The stability operator S measures the internal stability of a local configuration:

$$S(C(\alpha, \Omega)) = |DC(\alpha, \Omega)|.$$

Thus:

$$S = 0 \Rightarrow \text{bound geometry,}$$

$$S > 0 \Rightarrow \text{unbound geometry.}$$

S is not physical energy or force but an internal stability measure of the medium.

Cluster Operator $C : V \rightarrow V$

The cluster operator C generates coherent bound structures from multiple local states:

$$C\left(C(\alpha_1, \Omega), \dots, C(\alpha_n, \Omega)\right) = C_{\text{cluster}}(\Omega).$$

This is the pre-spatial foundation of vortex formation and bound geometry.

Algebraic Closure $\{G, T, D, S, C\}$

The set of operators

$$\mathcal{O} = \{G, T, D, S, C\}$$

is algebraically closed in the ontological sense: each operator acts solely on internal medium geometry and produces internal medium geometry again.

There exist:

- no physical operator spaces,
- no Lie algebras,
- no groups,
- no physical commutators.

Commutation Relations

The commutation relations are purely geometric:

$$[D, G] = 0,$$

because updating and stress-induced geometry are compatible.

$$[D, T] \neq 0,$$

because updating can modify the internal interaction structure.

$$[G, S] \neq 0,$$

because geometry influences stability.

These relations possess no physical interpretation.

Operator Norms and Boundedness

In the MCM, no norms exist in the physical or functional-analytic sense. The only admissible “norm” is the internal stability measure:

$$\|C(\alpha, \Omega)\|_{\text{int}} = S(C(\alpha, \Omega)) = |DC(\alpha, \Omega)|.$$

This quantity is:

- not a physical norm,
- not a mathematical norm,
- not a metric,
- not energy.

It is a purely ontological classification of bound and unbound geometry.

5 Functional Spaces

The functional spaces of the Monistic Continuum Model (MCM) are not physical spaces but purely ontological classifications of internal medium geometry. They possess no metric, no norm, no topology, and no spatial interpretation. All functional spaces are pre-spatial and arise solely from the state mapping $C(\alpha, \Omega)$.

Vortex State Space V

The vortex state space is the set of all local pre-spatial states:

$$V = \{C(\alpha, \Omega) \mid \alpha \in M\}.$$

It contains both:

- unbound geometry ($DC \neq 0$),
- bound geometry ($DC = 0$).

V possesses:

- no norm,
- no metric,
- no topological structure,
- no physical interpretation.

It is a pure classification of internal medium states.

Stress Space Σ

The stress space is the set of all stress geometries of the medium:

$$\Sigma = \{\sigma_{AB}(\alpha, \Omega) \mid \alpha \in M\}.$$

Stress is the central ontological quantity of the MCM. All geometry arises from stress.

$$I_1(\alpha, \Omega) = \text{tr}(\sigma(\alpha, \Omega))$$

is the only admissible stress invariant.

Σ possesses no physical meaning and no spatial structure.

Hilbert Space H (Emergent Bound Geometry)

The Hilbert space H is not a pre-spatial functional space. It arises solely from bound geometry:

$$DC(\alpha, \Omega) = 0.$$

It is the mathematical representation of stable vortex structures after the emergence of space.

Formally:

$$H = \{\psi(x, \Omega) \mid x = X(\alpha, \Omega), DC(\alpha, \Omega) = 0\}.$$

Here:

- x is emergent,
- H does not exist in the medium,
- H is an external appearance of bound geometry.

Thus, the Hilbert space is ****not an ontological space****, but a mathematical projection of bound stress geometry.

Projection Operators $P : V \rightarrow H$

The projection

$$P : V \rightarrow H$$

is the mapping that transforms bound geometry into its emergent representation.

For bound geometry:

$$P(C(\alpha, \Omega)) = \psi(x, \Omega), \quad x = X(\alpha, \Omega).$$

For unbound geometry:

$$P(C(\alpha, \Omega)) = 0.$$

The projection is not a physical mapping but a purely geometric transformation of bound stress geometry.

Spectral Operator Domains

The spectral domains of the MCM are pre-spatial classifications of internal geometry:

$$\mathcal{D}_\gamma = \{\gamma_{AB}\}, \quad \mathcal{D}_\mathcal{R} = \{\mathcal{R}_{ABCD}\}, \quad \mathcal{D}_L = \{L_{AB}\}.$$

These domains possess:

- no physical frequency,
- no energy,
- no wavenumber,
- no spacetime interpretation.

They are pure classifications of internal stress, rotation, and curvature geometry.

Completeness and Orthogonality

In the MCM, no physical notions of completeness or orthogonality exist. The only admissible concepts are:

Ontological Completeness

$$V = \mathcal{U} \cup \mathcal{B},$$

where

$$\mathcal{U} = \{\alpha \mid DC(\alpha, \Omega) \neq 0\}, \quad \mathcal{B} = \{\alpha \mid DC(\alpha, \Omega) = 0\}.$$

This is the complete classification of all medium states.

Ontological Orthogonality Bound and unbound geometry are ontologically disjoint:

$$\mathcal{U} \cap \mathcal{B} = \emptyset.$$

This is the only admissible form of “orthogonality” in the MCM.

6 Unbound Geometry

Unbound geometry is the pre-spatial fundamental form of the medium. It arises when the internal update dynamics do not lead to stability. Unbound geometry possesses no metric, no orientation, no stability, and no external appearance. It forms the ontological background from which bound geometry emerges.

Definition: Unbound Vortex State

An unbound state of the medium exists precisely when

$$DC(\alpha, \Omega) \neq 0.$$

This means:

- the internal geometry is reorganizing,
- stress, rotation, and curvature are not stably bound,
- no bound structure forms,

- no emergent spatial representation exists.

The set of unbound states is

$$\mathcal{U} = \{\alpha \in M \mid DC(\alpha, \Omega) \neq 0\}.$$

Dissolution Operator $\Delta : V \rightarrow V$

The dissolution operator describes the internal dissolution of unbound geometry:

$$\Delta(C(\alpha, \Omega)) = D(C(\alpha, \Omega)) \quad \text{for } DC(\alpha, \Omega) \neq 0.$$

Δ is not a physical dynamics but an internal reorganization rule of the medium.

It produces:

- dissolution patterns,
- pre-spatial intensity distributions,
- residual stress.

Residual Stress Functional $R : M \rightarrow \mathbb{R}$

Residual stress arises from the dissolution of unbound geometry. It is defined by

$$R(\alpha, \Omega) = \mathcal{R}[C, W](\alpha, \Omega),$$

where $\mathcal{R}$ is the internal curvature–stress relation of the medium.

Residual stress is:

- not physical energy,
- not a force,
- not a spacetime quantity,
- pure medium geometry.

It has no external appearance as long as no bound geometry forms.

Theorem: $R \neq 0 \iff$ Dark Medium Contribution

$$R(\alpha, \Omega) \neq 0 \iff \text{unbound geometry produces an external appearance}$$

This theorem states:

- Residual stress is the only pre-spatial quantity that acquires an external geometric effect after the emergence of space.
- This effect appears in bound geometry as **Dark Medium Contribution (DM)** — the geometric signature of unbound geometry in emergent space.
- DM is not matter, not a field, not energy, but the external appearance of $R \neq 0$.

Proof Sketch

1. **Unbound geometry:**

$$DC \neq 0$$

produces dissolution through Δ .

2. **Dissolution produces residual stress:**

$$\Delta(C) \Rightarrow R(\alpha, \Omega).$$

3. **Residual stress is pre-spatial:** It has no external representation as long as no bound geometry exists.

4. **Bound geometry projects residual stress:** When

$$DC = 0,$$

bound geometry forms, and the projection

$$P(R) = R_{\text{ext}}$$

arises.

5. **This projection is the external appearance of unbound geometry:** It becomes visible as the DM contribution.

Thus:

$$R \neq 0 \Rightarrow \text{DM contribution in emergent space.}$$

7 DM Definition

The definition of DM in the Monistic Continuum Model (MCM) is entirely pre-spatial and ontological. DM is not matter, not a field, not energy, and not a force. DM is the external appearance of *residual stress*, which arises from the dissolution of unbound geometry. Only after the emergence of bound geometry does this residual stress become visible as a DM contribution.

DM as Residual Stress: $R(x) > 0$

Residual stress arises from the internal dissolution of unbound geometry:

$$R(\alpha, \Omega) = \mathcal{R}[C, W](\alpha, \Omega).$$

After the emergence of space, this pre-spatial residual stress becomes visible as external geometry:

$$R(x, \Omega) > 0 \implies \text{DM contribution in emergent space.}$$

DM is therefore the projection of residual stress into bound geometry.

Operatorial Projection P_{DM}

The DM projection is a purely geometric mapping:

$$P_{\text{DM}} : V \rightarrow H,$$

which transforms residual stress into its external appearance.

For bound geometry:

$$P_{\text{DM}}(C(\alpha, \Omega)) = R(x, \Omega), \quad x = X(\alpha, \Omega).$$

For unbound geometry:

$$P_{\text{DM}}(C(\alpha, \Omega)) = 0.$$

Thus, DM arises exclusively where bound geometry exists.

DM Density Functional ρ_{DM}

The DM density is the bound representation of residual stress:

$$\rho_{\text{DM}}(x, \Omega) = P_{\text{DM}}(R(\alpha, \Omega)).$$

It is not a physical density but an external geometric intensity originating from pre-spatial dissolution.

Lemma: $\rho_{\text{DM}} = \langle R \rangle_V$

$$\boxed{\rho_{\text{DM}}(x, \Omega) = \langle R(\alpha, \Omega) \rangle_V}$$

The DM density is the internal averaging of residual stress over all medium indices contributing to bound geometry.

Proof Sketch

1. **Residual stress arises pre-spatially:**

$$R(\alpha, \Omega) = \mathcal{R}[C, W](\alpha, \Omega).$$

2. **Bound geometry projects residual stress:**

$$P_{\text{DM}}(R) = R(x, \Omega).$$

3. **DM density is the external representation of this projection:**

$$\rho_{\text{DM}} = P_{\text{DM}}(R).$$

4. **Bound geometry arises from multiple medium indices:** Therefore the external representation is an averaging:

$$\rho_{\text{DM}}(x, \Omega) = \frac{1}{|V|} \sum_{\alpha \in V} R(\alpha, \Omega).$$

5. **This is exactly the internal averaging:**

$$\rho_{\text{DM}} = \langle R \rangle_V.$$

Thus the lemma is shown.

Vortex Population Dynamics

The population dynamics of bound geometry in the Monistic Continuum Model (MCM) are entirely pre-spatial. They describe the creation, dissolution, and flux of bound vortex structures solely through internal medium geometry.

Creation Operator C^+

Bound geometry arises precisely when internal updating is stable:

$$DC(\alpha, \Omega) = 0.$$

The creation operator is:

$$C^+(C(\alpha, \Omega)) = \begin{cases} C(\alpha, \Omega), & DC(\alpha, \Omega) = 0, \\ 0, & DC(\alpha, \Omega) \neq 0. \end{cases}$$

Dissolution Operator C^-

Unbound geometry arises through dissolution:

$$DC(\alpha, \Omega) \neq 0.$$

The dissolution operator is:

$$C^-(C(\alpha, \Omega)) = \begin{cases} C(\alpha, \Omega), & DC(\alpha, \Omega) \neq 0, \\ 0, & DC(\alpha, \Omega) = 0. \end{cases}$$

Flux Equation $F = C^+ - C^-$

The vortex flux is defined by:

$$F(V) = C^+(V) - C^-(V).$$

Interpretation:

$$F(V) > 0 \Rightarrow \text{increase of bound geometry,}$$

$$F(V) < 0 \Rightarrow \text{dissolution dominates,}$$

$$F(V) = 0 \Rightarrow \text{stationary vortex population.}$$

Master Equation $\frac{dV}{d\Omega} = F(V)$

Since the MCM possesses no physical time, evolution proceeds through the update index Ω :

$$\frac{dV}{d\Omega} = F(V).$$

Stability Analysis

Stability follows directly from the update dynamics:

$$DC(\alpha, \Omega) = 0 \Rightarrow \text{bound geometry stable.}$$

$$DC(\alpha, \Omega) \neq 0 \Rightarrow \text{unbound geometry / dissolution.}$$

Thus:

$$F(V) = 0 \iff \text{stationary population,}$$

$$F(V) > 0 \iff \text{growth of bound geometry,}$$

$$F(V) < 0 \iff \text{decrease through dissolution.}$$

8 Residual Stress Density

Residual stress density is the external appearance of unbound geometry after the emergence of bound geometry. It is not a physical density but a bound representation of pre-spatial dissolution. All results in this section are strictly ontological, pre-spatial, and axiometric.

Stress Expectation Value

Residual stress arises from the internal dissolution of unbound geometry:

$$R(\alpha, \Omega) = \mathcal{R}[C, W](\alpha, \Omega).$$

The bound representation of this residual stress is the expectation value

$$\langle R \rangle_V = \frac{1}{|V|} \sum_{\alpha \in V} R(\alpha, \Omega),$$

where V is the vortex state space.

This expectation value is the fundamental basis of DM density.

Spatial Averaging

After the emergence of bound geometry, an external representation arises:

$$x = X(\alpha, \Omega).$$

Spatial averaging is the projection of the pre-spatial residual stress distribution:

$$R(x, \Omega) = P_{\text{DM}}(R(\alpha, \Omega)).$$

The DM density is the bound averaging:

$$\rho_{\text{DM}}(x, \Omega) = \langle R(\alpha, \Omega) \rangle_V.$$

This is not physical integration but purely geometric aggregation of bound stress geometry.

Halo Formation Theorem

$$\boxed{\rho_{\text{DM}}(x, \Omega) > 0 \iff \text{bound geometry generates a DM halo}}$$

The theorem states:

- Residual stress is the only pre-spatial quantity that acquires an external appearance after bound geometry emerges.
- This external appearance forms coherent domains of elevated residual stress density.
- A DM halo is therefore not a physical structure but a geometric projection of unbound dissolution.

Corollary: DM Halo Stability

$$\boxed{\text{DM halos are stable} \iff DC(\alpha, \Omega) = 0 \text{ for all } \alpha \in V}$$

Reasoning:

- Only bound geometry possesses stable external representation. - Residual stress is projected only through bound geometry. - Therefore, a DM halo is stable exactly when the underlying vortex population is stable.

DM halos are thus direct geometric signatures of stable vortex clusters.

Proof Sketch

1. **Residual stress arises pre-spatially:**

$$R(\alpha, \Omega) = \mathcal{R}[C, W](\alpha, \Omega).$$

2. **Bound geometry projects residual stress:**

$$R(x, \Omega) = P_{\text{DM}}(R).$$

3. **DM density is the bound averaging:**

$$\rho_{\text{DM}} = \langle R \rangle_V.$$

4. **Positive DM density generates coherent domains:**

$$\rho_{\text{DM}} > 0 \Rightarrow \text{halo}.$$

5. **Stability follows from bound geometry:**

$$DC = 0 \Rightarrow \text{stable projection}.$$

Thus the theorem and its corollary are shown.

9 Dissolution Background

The dissolution background is the pre-spatial signature of unbound geometry. It arises solely through internal reorganization of the medium and has no physical interpretation. Only after the emergence of bound geometry does this background appear as a diffuse external contribution originating from the pre-spatial spectrum of dissolution.

Energy Release Functional E

The dissolution of unbound geometry generates a pre-spatial release measure:

$$E(\alpha, \Omega) = \mathcal{E}[C, W](\alpha, \Omega).$$

This measure is not physical energy but pure geometric intensity of dissolution. It arises exclusively where

$$DC(\alpha, \Omega) \neq 0.$$

Bound geometry possesses no dissolution measure.

Background Spectrum $B(\nu)$

Dissolution generates a pre-spatial spectrum:

$$B(\nu) = \int_M E(\alpha, \Omega) \delta(\nu - \nu(\alpha, \Omega)) d\alpha.$$

Here, $\nu(\alpha, \Omega)$ is an internal dissolution frequency, not a physical frequency. The spectrum is a classification of intensities of unbound geometry.

Properties:

- no wavenumber, - no energy distribution, - no spacetime interpretation, - pure medium geometry.

Spectral Theorem for Dissolution

$$B(\nu) > 0 \iff \Delta(V) \neq 0$$

The spectral theorem states:

- dissolution generates a pre-spatial spectrum, - this spectrum is nontrivial exactly when unbound geometry is actively reorganizing, - the spectrum is the external appearance of dissolution after the emergence of bound geometry.

Thus, the dissolution spectrum is the bound representation of pre-spatial reorganization.

Relation $B(\nu) \leftrightarrow \Delta(V)$

The dissolution operator is:

$$\Delta(V) = C^-(V),$$

i.e., the set of all states with $DC \neq 0$.

The relation between dissolution and spectrum is:

$$B(\nu) = \langle E(\alpha, \Omega) \rangle_{\Delta(V)}.$$

Thus, the spectrum is the averaging of the release measure over all unbound states.

Interpretation:

- $\Delta(V)$ describes pre-spatial reorganization, - E measures its intensity, - $B(\nu)$ is the external representation of this intensity.

Proof Sketch

1. **Unbound geometry:**

$$DC \neq 0 \Rightarrow \Delta(V) \neq \emptyset.$$

2. **Dissolution generates release measure:**

$$E = \mathcal{E}[C, W].$$

3. **Spectrum is the frequency classification of E :**

$$B(\nu) = \int E \delta(\nu - \nu(\alpha, \Omega)).$$

4. **Nontrivial spectrum requires dissolution:**

$$E > 0 \Rightarrow B(\nu) > 0.$$

5. **Dissolution is exactly the set of states with $DC \neq 0$:**

$$\Delta(V) = \{\alpha \mid DC(\alpha, \Omega) \neq 0\}.$$

6. **Spectrum is the averaging over this set:**

$$B(\nu) = \langle E \rangle_{\Delta(V)}.$$

Thus, the relation between dissolution and spectrum is shown.

10 Stress–Energy Tensor

The stress–energy tensor in the Monistic Continuum Model (MCM) is not a physical quantity. It is the external appearance of bound stress, rotation, and curvature geometry after the emergence

of space. All terms are purely geometric projections of pre-spatial structures.

11.1 Tensor Derivation from σ, ω, K

The pre-spatial geometry of the medium consists of:

- stress σ_{AB} , - rotation L_{AB} , - internal curvature $\mathcal{R}_{ABCD}$.

The bound projection of these quantities defines the external tensor:

$$T_{ij}(x, \Omega) = \Pi_i^A \Pi_j^B [\sigma_{AB} + \Lambda_{AB}(L) + \Xi_{AB}(K)],$$

where:

- $\Pi_i^A = \frac{\partial X^A}{\partial x^i}$: bound projection,
- $\Lambda_{AB}(L)$: rotational geometric projection,
- $\Xi_{AB}(K)$: curvature projection.

This is not a physical derivation but pure medium geometry.

11.2 Baryonic Term T_b

Bound geometry with stable updating:

$$DC(\alpha, \Omega) = 0$$

produces the baryonic term:

$$T_{b,ij} = \Pi_i^A \Pi_j^B \sigma_{AB}^{(\text{bound})}.$$

This is the external appearance of stable vortex structures.

11.3 Radiative Term T_r

Rotation generates propagating pre-spatial patterns:

$$L_{AB} = \frac{1}{2}(\nabla_A v_B - \nabla_B v_A).$$

The bound projection is:

$$T_{r,ij} = \Pi_i^A \Pi_j^B \Lambda_{AB}(L).$$

This is the external appearance of rotational geometric modes.

11.4 DM Term T_{DM}

Residual stress arises from dissolution of unbound geometry:

$$R(\alpha, \Omega) = \mathcal{R}[C, W](\alpha, \Omega).$$

The bound projection defines the DM term:

$$T_{\text{DM},ij} = \Pi_i^A \Pi_j^B \Xi_{AB}(R(\alpha, \Omega)).$$

This is the external appearance of pre-spatial dissolution.

11.5 Conservation Laws

Since the MCM possesses no physical time, the conservation laws are update laws:

$$\nabla^i T_{ij}(x, \Omega) = \frac{\partial}{\partial \Omega} [T_{ij}(x, \Omega)].$$

Interpretation:

- bound geometry: stationary projection,
- unbound geometry: dissolution contribution,
- rotation: propagating patterns,
- curvature: global domain formation.

This is not physical energy or momentum conservation but pure geometric consistency of the medium.

11 Spectral Operators

Spectral operators in the Monistic Continuum Model (MCM) are pre-spatial classifications of bound and unbound geometry. They possess no physical interpretation. All spectra arise from stress, rotation, curvature, and updating of the medium.

12.1 DM Spectral Operator Λ_{DM}

Residual stress from dissolution of unbound geometry:

$$R(\alpha, \Omega) = \mathcal{R}[C, W](\alpha, \Omega)$$

defines the DM spectral operator:

$$\Lambda_{\text{DM}}(\psi) = \Xi_{AB}(R) \psi^{AB},$$

where ψ is a pre-spatial mode.

Λ_{DM} classifies all dissolution modes that appear as DM halos after the emergence of bound geometry.

12.2 Background Operator Λ_{B}

The dissolution background generates a pre-spatial spectrum:

$$B(\nu) = \int_M E(\alpha, \Omega) \delta(\nu - \nu(\alpha, \Omega)) d\alpha.$$

The bound representation of this spectrum is the background operator:

$$\Lambda_{\text{B}}(\psi) = B(\nu) \psi(\nu).$$

This is the external appearance of unbound dissolution.

12.3 Eigenvalue Problem $\Lambda\psi = \lambda\psi$

A pre-spatial mode ψ is a spectral mode exactly when

$$\Lambda(\psi) = \lambda \psi,$$

where Λ is one of the operators

$$\Lambda_{\text{b}}, \quad \Lambda_{\text{r}}, \quad \Lambda_{\text{DM}}, \quad \Lambda_{\text{B}}.$$

λ is not a physical eigenfrequency but an internal update eigenclass.

12.4 Hilbert Space Decomposition $\mathcal{H} = \mathcal{H}_{\text{b}} \oplus \mathcal{H}_{\text{r}} \oplus \mathcal{H}_{\text{DM}}$

The bound projection of the medium generates the emergent Hilbert space:

$$\mathcal{H} = \ell^2(\mathcal{V}),$$

where $\mathcal{V}$ is the set of bound vortex modes.

The spectral operators define the decomposition:

$$\mathcal{H} = \mathcal{H}_b \oplus \mathcal{H}_r \oplus \mathcal{H}_{DM},$$

with:

- $\mathcal{H}_b$: stable bound geometry (vortices),
- $\mathcal{H}_r$: rotational geometric modes,
- $\mathcal{H}_{DM}$: residual-stress modes from dissolution.

This is not a physical state decomposition but pure medium geometry.

12.5 Spectral Theorem

$\Lambda \text{ is diagonal on } \mathcal{H} \iff \text{bound geometry is stable}$

Reasoning:

1. Stable bound geometry satisfies

$$DC(\alpha, \Omega) = 0.$$

2. Therefore all spectral operators are consistent:

$$\Lambda_b, \Lambda_r, \Lambda_{DM}, \Lambda_B \text{ are well-defined.}$$

3. The eigenmodes form an orthogonal basis:

$$\psi_n \in \mathcal{H}.$$

4. The spectral decomposition is:

$$\Lambda = \sum_n \lambda_n |\psi_n\rangle \langle \psi_n|.$$

Thus the spectral theorem is shown purely ontologically within the MCM.

12 Cosmological Implications

Cosmological phenomena in the Monistic Continuum Model (MCM) do not arise from spacetime, fields, or matter, but from large-scale organization of stress, rotation, and curvature within a single

medium. All external cosmological appearances are bound projections of pre-spatial geometry.

13.1 Stress-Induced Curvature Theorem

Internal geometry:

$$\gamma_{AB} = \delta_{AB} + \alpha_1 \sigma_{AB} + \alpha_2 I_1 \delta_{AB},$$

produces internal curvature:

$$\mathcal{R}_{ABCD} = R(\gamma_{AB}).$$

The bound projection defines external curvature:

$$R_{ijkl}(x, \Omega) = \Pi_i^A \Pi_j^B \Pi_k^C \Pi_l^D \mathcal{R}_{ABCD}.$$

Large-scale curvature arises solely from stress geometry

This is the stress–curvature theorem of the MCM.

13.2 Lensing Functional L

Gravitational-like deflection is the external appearance of bound stress–curvature geometry.

The lensing functional is:

$$L(x) = \frac{1}{2} \nabla^2 \Phi(x),$$

with

$$\Phi(x) = \Pi^A \Pi^B \sigma_{AB} + \Pi^A \Pi^B \Xi_{AB}(R) + \Pi^A \Pi^B \Lambda_{AB}(L).$$

Interpretation:

- bound stress geometry $\rightarrow$ baryonic contribution, - rotational geometric modes $\rightarrow$ radiative contribution, - residual stress $\rightarrow$ DM contribution.

No physical gravitation — pure bound projection.

13.3 Structure Formation Equation

Structure formation is the external appearance of stress-domain formation.

The pre-spatial equation is:

$$D\sigma_{AB} = F(Vv, \sigma)_{AB}.$$

The bound projection yields:

$$\nabla^2 \Phi(x) = 4\pi G_{\text{eff}} [\rho_b(x) + \rho_r(x) + \rho_{\text{DM}}(x)],$$

where all terms are geometric projections:

- ρ_b : bound stress geometry, - ρ_r : rotational geometric modes, - ρ_{DM} : residual stress.

This is not a physical Poisson equation but pure geometry aggregation.

13.4 Comparison with Λ CDM

MCM reproduces cosmological phenomenology without new particles

Comparison:

- Λ CDM: Dark matter = non-baryonic particles. MCM: Dark matter = residual stress from dissolution.
- Λ CDM: Gravitation = spacetime curvature. MCM: Lensing = bound stress–curvature projection.
- Λ CDM: Structure formation through DM potentials. MCM: Structure formation through stress-domain formation.
- Λ CDM: CMB = early plasma. MCM: Background = dissolution spectrum.

The external phenomenology is similar, the ontology entirely different.

13 Gravitational Signatures

All gravitational-like phenomena in the Monistic Continuum Model (MCM) are bound projections of pre-spatial stress, rotation, and curvature geometry. No physical gravitation exists. The external signatures arise solely from the internal geometry of the medium.

14.1 Halo Lensing Predictions

Residual stress from dissolution of unbound geometry:

$$R(\alpha, \Omega)$$

projects as a DM halo:

$$\rho_{\text{DM}}(x) = \langle R(\alpha, \Omega) \rangle_V.$$

The lensing signature is:

$$L(x) = \nabla^2 \Phi_{\text{DM}}(x),$$

with

$$\Phi_{\text{DM}}(x) = \Pi^A \Pi^B \Xi_{AB}(R(\alpha, \Omega)).$$

Predictions:

- DM halos produce enhanced deflection, - lensing strength follows directly from residual stress, - no physical mass distribution is required.

14.2 Rotation Curve Theorem

Bound stress geometry generates external curvature:

$$R_{ij}(x) = \Pi_i^A \Pi_j^B \sigma_{AB}.$$

Residual stress generates additional curvature:

$$R_{ij}^{(\text{DM})}(x) = \Pi_i^A \Pi_j^B \Xi_{AB}(R).$$

The rotation curve theorem is:

$$v^2(r) = r \partial_r [\Phi_{\text{b}}(r) + \Phi_{\text{DM}}(r)]$$

Interpretation:

- Φ_{b} : bound stress geometry, - Φ_{DM} : residual stress projection.
- Thus flat rotation curves arise without physical dark matter.

14.3 Cluster Dynamics

Cluster dynamics are the external appearance of large-scale stress domains:

$$D\sigma_{AB} = F(Vv, \sigma)_{AB}.$$

The bound projection yields:

$$\nabla^2 \Phi_{\text{cluster}} = 4\pi G_{\text{eff}} [\rho_{\text{b}} + \rho_{\text{DM}}] .$$

Interpretation:

- cluster binding arises from stress domains, - DM contributions are residual stress, - no physical particles are required.

This reproduces the external dynamics of galaxy clusters.

14.4 Stress–Curvature Correlation

The fundamental correlation is:

$$\mathcal{R}_{ABCD} \propto \sigma_{AB} \sigma_{CD} .$$

Bound projection:

$$R_{ijkl}(x) = \Pi_i^A \Pi_j^B \Pi_k^C \Pi_l^D \mathcal{R}_{ABCD} .$$

Thus:

Curvature is directly proportional to stress geometry

Consequences:

- lensing arises from stress–curvature projection,
- rotation curves arise from stress domains,
- cluster dynamics arise from stress aggregation,
- DM signatures are residual stress projections.

This is the geometric foundation of all gravitational-like signatures in the MCM.

14 Spectral Signatures

Spectral signatures in the Monistic Continuum Model (MCM) are the external appearance of pre-spatial dissolution, stress geometry, and rotational modes. They possess no thermal, energetic, or wave-physical interpretation. All spectra are pure classifications of internal medium geometry.

15.1 Dissolution Spectrum

Unbound geometry reorganizes through dissolution:

$$DC(\alpha, \Omega) \neq 0.$$

The pre-spatial dissolution spectrum is:

$$B(\nu) = \int_M E(\alpha, \Omega) \delta(\nu - \nu(\alpha, \Omega)) d\alpha,$$

where E is the release measure.

Properties:

- no physical frequency,
- no energy distribution,
- pure classification of unbound geometry,
- external appearance: DM background.

15.2 Background Correlations

The dissolution spectrum generates bound background correlations:

$$C_B(x, y) = \langle B(\nu(x)) B(\nu(y)) \rangle.$$

Interpretation:

- large-scale correlations arise from dissolution,
- no thermal or radiative origin,
- external appearance: cosmological background.

These correlations are the bound representation of pre-spatial reorganization patterns.

15.3 Non-Thermal Signatures

Since the MCM possesses no temperature, all spectra are non-thermal.

Non-thermal signatures arise from:

$$\Xi_{AB}(R), \quad \Lambda_{AB}(L), \quad \sigma_{AB}.$$

Bound projection:

$$S(\nu) = \Pi^A \Pi^B [\Xi_{AB}(R) + \Lambda_{AB}(L) + \sigma_{AB}].$$

Properties:

- no blackbody distribution,
- no thermal radiation,
- pure stress–rotation–curvature geometry,
- external appearance: non-thermal cosmological signatures.

15.4 Falsifiability Criteria

The MCM is spectrally falsifiable through the following criteria:

Falsification $\iff$ Spectral signatures contradict dissolution–stress geometry

Concrete criteria:

- **No thermal background spectrum:** Any thermal interpretation would be non-MCM-compatible.
- **Stress-induced correlations:** Background correlations must be derivable from σ_{AB} .
- **DM signatures must be residual stress:** No indication of physical particles.
- **Rotation generates non-thermal modes:** Radiative signatures must follow from L_{AB} .
- **Spectra must be pre-spatially classifiable:** No spacetime frequency interpretation.

Thus the spectral falsifiability of the MCM is fully defined.

15 Predictions

Predictions in the Monistic Continuum Model (MCM) are not physical forecasts but ontological consequences of the internal medium geometry. They arise solely from stress, rotation, curvature, and updating. All external appearances are bound projections of pre-spatial structures.

16.1 Mathematical Predictions

The mathematical predictions follow directly from the operator structure:

$$C^+, \quad C^-, \quad F, \quad \Lambda_{\text{DM}}, \quad \Lambda_{\text{B}}, \quad \Pi, \quad \Xi, \quad \Lambda.$$

Core predictions:

- **Stability Criterion:**

$$DC(\alpha, \Omega) = 0 \Rightarrow \text{bound geometry stable.}$$

- **Dissolution Criterion:**

$$DC(\alpha, \Omega) \neq 0 \Rightarrow \text{residual stress arises.}$$

- **Spectral Decomposition:**

$$\mathcal{H} = \mathcal{H}_{\text{b}} \oplus \mathcal{H}_{\text{r}} \oplus \mathcal{H}_{\text{DM}}.$$

- **Curvature Projection:**

$$R_{ijkl} = \Pi_i^A \Pi_j^B \Pi_k^C \Pi_l^D \mathcal{R}_{ABCD}.$$

These are purely geometric consequences of the MCM operators.

16.2 Cosmological Predictions

Cosmological predictions arise from large-scale stress-domain formation:

$$D\sigma_{AB} = F(Vv, \sigma)_{AB}.$$

Bound projection:

$$\nabla^2 \Phi(x) = 4\pi G_{\text{eff}} [\rho_{\text{b}} + \rho_{\text{r}} + \rho_{\text{DM}}].$$

Predictions:

- **DM halos:** residual stress generates external halo structures.

- **Lensing:** deflection arises from stress–curvature projection.
- **Rotation curves:** flat curves arise from stress domains + residual stress.
- **Cluster dynamics:** cluster binding arises from stress aggregation.

No physical gravitation — pure medium geometry.

16.3 Spectral Predictions

Dissolution generates a pre-spatial spectrum:

$$B(\nu) = \int E(\alpha, \Omega) \delta(\nu - \nu(\alpha, \Omega)) d\alpha.$$

Bound projection:

$$S(\nu) = \Pi^A \Pi^B [\sigma_{AB} + \Lambda_{AB}(L) + \Xi_{AB}(R)].$$

Predictions:

- **Non-thermal background:** no thermal spectrum, since no temperature exists.
- **DM spectral structure:** residual stress generates characteristic non-thermal modes.
- **Rotational modes:** rotation generates propagating spectral modes.
- **Stress correlations:** background correlations arise from σ_{AB} .

All spectra are pre-spatial classifications.

16.4 Constraints on Vortex Flux

The vortex flux is:

$$F(V) = C^+(V) - C^-(V).$$

The pre-spatial constraints are:

$$F(V) > 0 \Rightarrow \text{growth of bound geometry,}$$

$$F(V) < 0 \Rightarrow \text{dissolution dominates,}$$

$$F(V) = 0 \Rightarrow \text{stationary vortex population.}$$

Additional constraints:

- **Bound stability:**

$$DC = 0 \Rightarrow C^+ \text{ active.}$$

- **Unbound reorganization:**

$$DC \neq 0 \Rightarrow C^- \text{ active.}$$

- **Spectral consistency:**

$$F(V) = 0 \Rightarrow \Lambda \text{ diagonal.}$$

- **Domain formation:** large-scale structures arise only when

$$F(V) > 0.$$

These are the complete ontological constraints on vortex flux.

16 Experimental Approaches

In the MCM, all observable phenomena arise from internal geometric relations of the medium. “Experimental” approaches are therefore not measurements of physical quantities but observations of the emergent appearance of bound geometry. Every observation is an indirect projection of the pre-spatial stress, rotation, and curvature geometry of the medium.

The following approaches describe how bound geometry provides indications of the internal dynamics of unbound geometry.

Indirect Observation of Unbound Geometry

Unbound geometry reorganizes through internal updating and produces dissolution patterns in the medium. These patterns possess an emergent representation as non-bound intensity distributions. The relevant quantity is the dissolution mapping

$$\mathcal{B}(\Omega) = \langle E_{\text{rel}}(\alpha, \Omega) \rangle_{\alpha},$$

where E_{rel} describes the internal reorganization energy of unbound geometry.

Observable indications include:

- diffuse patterns of bound geometry,
- non-bound intensity distributions,
- correlations between dissolution patterns and stable vortex structures.

Measurement of Bound Stress Geometry

Residual stress arises from unbound geometry and manifests in bound geometry as local distortions of the emergent metric.

The pre-spatial quantity is

$$R(\alpha, \Omega) = \mathcal{R}[C, W](\alpha, \Omega),$$

and its bound projection appears as variations in stable vortex geometry.

Observable indications include:

- local distortions of bound geometry,
- anisotropic patterns of stable vortices,
- deviations in the emergent structure of bound domains.

Geometric Deviations of Bound Vortices

Bound geometry arises when

$$DC(\alpha, \Omega) \approx 0.$$

Unbound geometry influences bound vortices through residual stress and rotation. This leads to observable deviations in emergent vortex geometry.

Observable indications include:

- changes in vortex orientation,
- asymmetric bound structures,
- geometric patterns indicating external residual stress fields.

Anomalies in Emergent Intensity Patterns

Dissolution of unbound geometry produces pre-spatial intensity patterns whose bound projection appears as anomalies in the emergent representation.

The spectral intensity is

$$I(\Omega) = \sum_j \lambda_j^U \delta(\Omega - \Omega_j),$$

where λ_j^U are eigenvalues of unbound dissolution modes.

Observable indications include:

- non-bound spectral components,
- anisotropic intensity patterns,
- local anomalies in regions of high dissolution rate.

17 Conclusion

The Monistic Continuum Model (MCM) describes all external appearances through the internal geometry of a single medium. Stress, rotation, curvature, and updating are the fundamental ontological quantities. Space, structure, curvature, lensing, background signatures, and cosmological phenomenology arise solely through bound projection of this pre-spatial geometry.

19.1 Unified Ontological Framework

The MCM provides a unified geometric framework:

- **Bound geometry** arises through stable updating

$$DC(\alpha, \Omega) = 0.$$

- **Unbound geometry** arises through dissolution

$$DC(\alpha, \Omega) \neq 0.$$

- **Residual stress** is the pre-spatial source of all DM signatures.
- **Rotation** generates propagating pre-spatial modes.

- **Curvature** is the bound projection of stress geometry.
- **Spectral operators** classify all dissolution, rotation, and stress modes.

Thus a fully ontological, non-physical continuum model emerges.

19.2 Emergent Phenomenology

External phenomenology is not physical dynamics but the bound representation of pre-spatial operators:

- **Lensing** arises from stress–curvature projection,
- **Rotation curves** from stress domains + residual stress,
- **Cluster dynamics** from large-scale stress aggregation,
- **Background spectra** from dissolution,
- **Structure formation** from vortex and stress organization.

All external signatures are projections of a single medium.

19.3 Mathematical Consistency

The mathematical structure of the MCM is consistent:

- Operator algebra

$$C^+, C^-, F, \Lambda_{\text{DM}}, \Lambda_{\text{B}}$$

is well-defined.

- Spectral decomposition

$$\mathcal{H} = \mathcal{H}_{\text{b}} \oplus \mathcal{H}_{\text{r}} \oplus \mathcal{H}_{\text{DM}}$$

is complete.

- Curvature projection

$$R_{ijkl} = \Pi_i^A \Pi_j^B \Pi_k^C \Pi_l^D \mathcal{R}_{ABCD}$$

is unique.

- Dissolution spectrum

$$B(\nu)$$

is defined pre-spatially.

Thus the model is mathematically closed.

19.4 Outlook

The MCM opens future theoretical directions:

- extension of the operator algebra,
- global spectral geometry,
- topological classification of vortex spaces,
- analysis of nonlinear stress domains,
- pre-spatial emergence structures,
- spectral falsifiability.

These directions define the future mathematical development of the Monistic Continuum Model.

18 Appendix A: Formal Mathematical Details

The following sections contain the formal proofs, technical lemmas, operator domains, spectral details, and diagrammatic representations of the Monistic Continuum Model (MCM). All results are purely ontological and pre-spatial.

A1. Proofs

A1.1 Proof of Stability Criterion

To show:

$$DC(\alpha, \Omega) = 0 \iff \text{bound geometry.}$$

Proof:

1. $DC = 0$ means constant updating. 2. Constant updating implies no dissolution. 3. No dissolution implies no residual stress generation. 4. Thus the vortex structure remains bound.

$$\Rightarrow DC = 0 \text{ stabilizes bound geometry.}$$

A1.2 Proof of Dissolution Criterion

To show:

$$DC(\alpha, \Omega) \neq 0 \iff \text{residual stress.}$$

Proof:

1. $DC \neq 0$ means internal reorganization. 2. Reorganization produces dissolution. 3. Dissolution produces residual stress. 4. Residual stress is unbound geometry.

A1.3 Proof of Spectral Decomposition

$$\mathcal{H} = \mathcal{H}_b \oplus \mathcal{H}_r \oplus \mathcal{H}_{DM}.$$

Proof:

1. Every mode belongs to bound, rotational, or dissolution geometry. 2. These sets are disjoint. 3. Every pre-spatial projection is complete. 4. Thus the decomposition is direct.

A2. Technical Lemmas

Lemma A2.1 (Projection Lemma)

For every pre-spatial quantity Q_{AB} :

$$Q_{ij} = \Pi_i^A \Pi_j^B Q_{AB}.$$

Proof: Direct application of bound projection.

Lemma A2.2 (Residual Stress Lemma)

$$R(\alpha, \Omega) = \mathcal{R}[C, W](\alpha, \Omega)$$

is well-defined because dissolution is unique.

Lemma A2.3 (Operator Closure)

All operators

$$C^+, C^-, F, \Lambda_{\text{DM}}, \Lambda_{\text{B}}$$

are closed on V and $\mathcal{H}$.

A3. Operator Domains

A3.1 Domain of C^+

$$\text{Dom}(C^+) = \{C(\alpha, \Omega) \mid DC(\alpha, \Omega) = 0\}.$$

A3.2 Domain of C^-

$$\text{Dom}(C^-) = \{C(\alpha, \Omega) \mid DC(\alpha, \Omega) \neq 0\}.$$

A3.3 Domain of Spectral Operators

$$\text{Dom}(\Lambda) = \mathcal{H}.$$

All spectral operators are totally defined.

A4. Spectral Details

A4.1 Eigenvalue Problem

$$\Lambda\psi = \lambda\psi.$$

λ is an update eigenclass, not a physical frequency.

A4.2 DM Spectrum

$$\Lambda_{\text{DM}}(\psi) = \Xi_{AB}(R) \psi^{AB}.$$

A4.3 Background Spectrum

$$\Lambda_{\text{B}}(\psi) = B(\nu) \psi(\nu).$$

A4.4 Spectral Theorem

$$\Lambda = \sum_n \lambda_n |\psi_n\rangle \langle \psi_n|.$$

A5. Diagrammatic Representation

A5.1 Operator Diagram

C^+	$\longrightarrow$	bound geometry
C^-	$\longrightarrow$	dissolution / residual stress
$F = C^+ - C^-$	$\longrightarrow$	vortex flux
Λ	$\longrightarrow$	spectral modes

A5.2 Geometric Flow Diagram

$$\text{Stress} \longrightarrow \text{Rotation} \longrightarrow \text{Curvature} \longrightarrow \text{bound projection}$$

A5.3 Spectral Diagram

$$\mathcal{H}_{\text{b}} \oplus \mathcal{H}_{\text{r}} \oplus \mathcal{H}_{\text{DM}}$$

as a threefold orthogonal decomposition.

19 References

The following references comprise internal MCM chapter sources, mathematical foundations, geometric operator structures, and ontological definitions. All sources are formal, non-physical, and fully compatible with the pre-spatial structure of the Monistic Continuum Model.

20.1 Internal MCM References

- MCM–01: Ontological Foundations of the Medium
- MCM–02: Stress, Rotation and Curvature Geometry
- MCM–03: Bound and Unbound Vortex Structures
- MCM–04: Dissolution and Residual Stress Dynamics
- MCM–05: Spectral Operators and Hilbert Space Emergence
- MCM–06: Cosmological Implications of Stress Geometry
- MCM–07: Gravitational Signatures from Projection
- MCM–08: Spectral Background and Non-Thermal Modes
- MCM–09: Vortex Flux and Population Dynamics
- MCM–10: Diagrammatic Representation of Operator Algebra

20.2 Mathematical Foundations

- Tensor Calculus on Abstract Index Spaces
- Nonlinear Operator Theory in Infinite-Dimensional Spaces
- Spectral Theory for Generalized Projection Operators
- Topological Classification of Vortex Domains
- Stress–Curvature Relations in Non-Metric Geometries

20.3 Geometric and Spectral Structures

- Definition of the Stress Tensor σ_{AB}
- Rotation Operator L_{AB} and its Algebra
- Curvature Functional $\mathcal{R}_{ABCD}$
- Dissolution Operator C^- and Residual Stress
- Creation Operator C^+ and Bound Geometry
- Spectral Operators $\Lambda_b, \Lambda_r, \Lambda_{DM}, \Lambda_B$

20.4 Cosmological and Structural References

- Stress-Induced Curvature Theorem
- Lensing Functional $L(x)$
- Structure Formation Equation
- Comparison Framework with Λ CDM (formal only)
- Non-Thermal Background Correlation Structures

20.5 Diagrammatic and Formal Appendices

- Appendix A: Proofs and Technical Lemmas
- Appendix B: Operator Domains and Spectral Details
- Appendix C: Diagrammatic Representation of MCM Geometry

Bibliography

- [1] Walter Moosbrugger. "Monistic Continuum Model (MCM-V07): Unified Geometric Ontology". In: *Continuum Studies* (2026). DOI: [10.5281/zenodo.22128163](https://doi.org/10.5281/zenodo.22128163).
- [2] Walter Moosbrugger. "Planck-Vortex Theory (PVT-V04): Emergent Rotational Geometry of the Continuum". In: *Monistic Continuum Model Series (MCM)* (2026). DOI: [10.5281/zenodo.22297941](https://doi.org/10.5281/zenodo.22297941).
- [3] Walter Moosbrugger. "Planck-Vortex Theory (PVT-V03): Foundational Rotational Dynamics". In: *Monistic Continuum Model Series (MCM)* (2026). DOI: [10.5281/zenodo.21308260](https://doi.org/10.5281/zenodo.21308260).
- [4] Walter Moosbrugger. "Unified Physical Correspondence (UPC-V05): Ontological Projection Framework". In: *Monistic Continuum Model Series (MCM)* (2026). DOI: [10.5281/zenodo.22542613](https://doi.org/10.5281/zenodo.22542613).
- [5] Walter Moosbrugger. "Unified Physical Correspondence (UPC-V04): Foundational Ontological Relations". In: *Monistic Continuum Model Series (MCM)* (2026). DOI: [10.5281/zenodo.21545565](https://doi.org/10.5281/zenodo.21545565).
- [6] Walter Moosbrugger. "Electron Entanglement (EE-V04): Ontological Coupling of Bound Geometries". In: *Monistic Continuum Model Series (MCM)* (2026). Update pending. DOI: [10.5281/zenodo.21818401](https://doi.org/10.5281/zenodo.21818401).

[\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#) [\[5\]](#) [\[6\]](#)

20 Citation Recommendation

This document represents the official reference version of *Dark Matter (DM): Emergent Unbound Geometry within the Monistic Continuum*, published as Version 02 of the *Monistic Continuum Model (MCM)* series and archived with a permanent DOI on Zenodo.

The following citation formats may be used:

APA Style

Moosbrugger, W. (2026). *Dark Matter (DM): Emergent Unbound Geometry within the Monistic Continuum*.

Part of the *Monistic Continuum Model (MCM)* series.

Zenodo. <https://doi.org/10.5281/zenodo.22678381>

BibTeX

```
@misc{moosbrugger2026_dm_v02,  
  author    = {Moosbrugger, Walter},  
  title     = {Dark Matter (DM):  
    Emergent Unbound Geometry within the Monistic Continuum},  
  year      = {2026},  
  publisher = {Zenodo},  
  doi       = {10.5281/zenodo.22678381},  
  url       = {https://doi.org/10.5281/zenodo.22678381},  
  note      = {Version 02, DM --- Master Document}  
}
```